

PRODUCT AND CATEGORY MANAGEMENT

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	INT	Primary Key
Category_Name	Varchar(100)	NOT NULL UNIQUE
Description	Varchar(200)	-

1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	INT	Primary Key
Product_Name	Varchar(100)	NOT NULL
Seller_ID	INT	Foreign Key
Category_ID	INT	Foreign Key
Price	Number(10,2)	NOT NULL
Brand	Varchar(100)	-
Stock	INT	NOT NULL

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing

Category.Category_ID

- **Relationship:** One-to-Many (1:M)

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID INT PRIMARY KEY,  
3     Category_Name VARCHAR(100) UNIQUE,  
4     Description VARCHAR(255)  
5 );
```

Table created.

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID INT PRIMARY KEY,  
3     Product_Name VARCHAR(150) NOT NULL,  
4     Seller_ID INT,  
5     Category_ID INT,  
6     Price DECIMAL(10,2),  
7     Brand VARCHAR(100),  
8     Stock INT NOT NULL,  
9     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
10 );
```

Table created.

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)  
2 VALUES (1, 'Mobiles', 'Smartphones and mobile accessories');  
1 row created.  
  
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)  
2 VALUES (2, 'Laptops', 'Laptops and computer accessories');  
1 row created.  
  
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)  
2 VALUES (3, 'Fashion', 'Men and women clothing and fashion');  
1 row created.  
  
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)  
2 VALUES (4, 'Electronics', 'Electronic devices and accessories');  
1 row created.  
  
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)  
2 VALUES (5, 'Home Appliances', 'Appliances and products for home');  
1 row created.  
  
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)  
2 VALUES (6, 'Beauty', 'Beauty and personal care products');  
1 row created.  
  
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)  
2 VALUES (7, 'Books', 'Educational and general books');  
1 row created.  
  
SQL> INSERT INTO Category (Category_ID, Category_Name, Description)  
2 VALUES (8, 'Grocery', 'Food and daily grocery products');  
1 row created.
```

4.2 Product Records

```
SQL> INSERT INTO Product
2 (Product_ID, Product_Name, Seller_ID, Category_ID, Price, Brand, Stock)
3 VALUES
4 (110, 'Sony Wireless Headphones', 3, 4, 4999.00, 'Sony', 25);

row created.

SQL>
SQL> INSERT INTO Product
2 (Product_ID, Product_Name, Seller_ID, Category_ID, Price, Brand, Stock)
3 VALUES
4 (111, 'LG Washing Machine', 1, 5, 32999.00, 'LG', 10);

row created.

SQL>
SQL> INSERT INTO Product
2 (Product_ID, Product_Name, Seller_ID, Category_ID, Price, Brand, Stock)
3 VALUES
4 (112, 'Philips Mixer Grinder', 4, 5, 3499.00, 'Philips', 30);

row created.

SQL>
SQL> INSERT INTO Product
2 (Product_ID, Product_Name, Seller_ID, Category_ID, Price, Brand, Stock)
3 VALUES
4 (113, 'Lakme Face Cream', 5, 6, 499.00, 'Lakme', 50);

row created.

SQL>
SQL> INSERT INTO Product
2 (Product_ID, Product_Name, Seller_ID, Category_ID, Price, Brand, Stock)
3 VALUES
4 (114, 'The Alchemist', 6, 7, 299.00, 'Penguin', 40);

row created.

SQL>
SQL> INSERT INTO Product
2 (Product_ID, Product_Name, Seller_ID, Category_ID, Price, Brand, Stock)
3 VALUES
4 (115, 'Tata Salt 1kg', 7, 8, 30.00, 'Tata', 200);

row created.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL>
SQL> SELECT * FROM Category;

CATEGORY_ID
CATEGORY_NAME
DESCRIPTION
-----
1
mobiles
smartphones and mobile accessories
2
laptops
laptops and computer accessories
CATEGORY_ID
CATEGORY_NAME
DESCRIPTION
-----
3
fashion
men and women clothing and fashion
4
electronics
CATEGORY_ID
CATEGORY_NAME
DESCRIPTION
-----
5
home Appliances
appliances and products for home
6
CATEGORY_ID
CATEGORY_NAME
DESCRIPTION
-----
7
beauty
beauty and personal care products
8
books
```

5.2 Display Product Records

```
SQL> SELECT * FROM Product;
PRODUCT_ID
PRODUCT_NAME
SELLER_ID CATEGORY_ID PRICE
BRAND
STOCK
101
iPhone 15 1 69999
PRODUCT_ID
PRODUCT_NAME
SELLER_ID CATEGORY_ID PRICE
BRAND
STOCK
Apple 25
PRODUCT_ID
PRODUCT_NAME
SELLER_ID CATEGORY_ID PRICE
BRAND
STOCK
102
Samsung Galaxy S24 2 64999
PRODUCT_ID
PRODUCT_NAME
SELLER_ID CATEGORY_ID PRICE
BRAND
STOCK
Samsung 30
```

6. Product Update

```
SQL> UPDATE Product
2 SET Price = 67999
3 WHERE Product_ID = 101;

1 row updated.

SQL> UPDATE Product
2 SET Stock = 50
3 WHERE Product_ID = 101;

1 row updated.

SQL> UPDATE Product
2 SET Brand = 'Apple Inc.'
3 WHERE Product_ID = 101;

1 row updated.

SQL> UPDATE Product
2 SET Price = 65999,
3 Stock = 45
4 WHERE Product_ID = 102;

1 row updated.
```

7. Product Deletion

```
SQL> DELETE FROM Product
      2 WHERE Product_ID = 102;
```

1 row deleted.

```
SQL>
SQL> DELETE FROM Product
      2 WHERE Product_ID = 103;
```

1 row deleted.

```
SQL>
SQL> DELETE FROM Product
      2 WHERE Product_ID = 104;
```

1 row deleted.

```
SQL>
SQL> DELETE FROM Product
      2 WHERE Product_ID = 105;
```

1 row deleted.

8. Category-wise Product Reports

```
SQL> SELECT
      2 C.Category_ID,
      3 C.Category_Name,
      4 P.Product_ID,
      5 P.Product_Name,
      6 P.Brand,
      7 P.Price,
      8 P.Stock
      9 FROM Category C
     10 JOIN Product P
     11 ON C.Category_ID = P.Category_ID
     12 ORDER BY C.Category_ID;
```

CATEGORY_ID

CATEGORY_NAME

PRODUCT_ID

PRODUCT NAME

BRAND

PRICE STOCK

1

CATEGORY_ID

CATEGORY NAME

PRODUCT_ID

PRODUCT NAME

BRAND

PRICE STOCK

Mobiles

CATEGORY_ID

CATEGORY NAME

PRODUCT_ID

PRODUCT NAME

BRAND

PRICE STOCK

101

CATEGORY_ID

8.1 Product Count by Category

```
SQL> SQL> SELECT
  2     C.Category_ID,
  3     C.Category_Name,
  4     COUNT(P.Product_ID) AS Product_Count
  5 FROM Category C
  6 LEFT JOIN Product P
  7 ON C.Category_ID = P.Category_ID
  8 GROUP BY C.Category_ID, C.Category_Name
  9 ORDER BY C.Category_ID;
```

```
CATEGORY_ID
-----
CATEGORY_NAME
-----
PRODUCT_COUNT
-----
      1
Mobiles      1
      2
Laptops      0
CATEGORY_ID
-----
CATEGORY_NAME
-----
PRODUCT_COUNT
-----
      3
Fashion      2
      4
Electronics
CATEGORY_ID
-----
CATEGORY_NAME
-----
PRODUCT_COUNT
-----
      5
Home Appliances 2
      6
```